SafeHire: An AI-Powered Service Hiring Platform
Project Demonstration Report

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I. Project Overview

SafeHire is a web-based service hiring platform designed to connect employers with verified workers in a secure and efficient way. The system addresses common issues in traditional hiring such as lack of verification, poor matching, and absence of transparency. The platform integrates structured worker data, AI-assisted matching, and a complete hiring workflow including job posting, application, hiring, payment, and review.

II. System Objectives

The main objectives of SafeHire are to ensure worker authenticity through verification, provide intelligent job-worker matching, create a transparent hiring and transaction system, and improve trust in service-based digital platforms.

III. System Features and Implementation

A. Worker Management System

A structured worker registration system was developed where workers can create profiles with skills, experience, and address information. They can also provide identity details such as NID and phone number, and view job recommendations generated by the system. Admin verification ensures that only trusted workers are eligible for hiring.

B. AI-Based Job Matching System

A lightweight AI-based matching system was implemented to recommend jobs to workers and workers to employers. The system calculates a match score based on skill similarity, job category, title keyword matching, location similarity, experience, and worker rating. This approach improves hiring efficiency and reduces manual selection effort.

C. Job Posting and Application System

Employers can post jobs with title, category, location, and budget information, view job listings, and receive applications from workers. On the other hand, workers can view AI-matched jobs and apply for suitable opportunities through the platform.

D. Hiring Workflow System

A complete hiring workflow was developed in which job status changes from “Open” to “Assigned.” Employers can select workers based on AI ranking, and only verified workers are allowed to be hired. This ensures both security and efficiency in the hiring process.

E. Transaction Management System

A structured transaction system was implemented to track employer information, worker information, job details, payment amount, and transaction status. This feature ensures transparency and accountability throughout the service hiring process.

F. Payment Simulation System

A simulated “Pay Now” feature was added to demonstrate the financial workflow of the system. It updates transaction status from “Assigned” to “Completed” and allows full project demonstration without integrating a real payment gateway.

G. Review and Rating System

After job completion, employers can rate workers and provide reviews. Decimal ratings are supported, and workers receive average ratings based on completed tasks. This feature improves platform credibility and helps future employers make informed decisions.

H. Admin Dashboard and Monitoring

Administrative features include worker verification with status categories such as Pending, Verified, and Rejected. The admin can also view all jobs and workers, monitor transactions, and observe dashboard statistics for overall system management.

I. User Interface Design

The frontend was designed using Tailwind CSS with a dark theme and responsive layout. Key user interface elements include clean dashboards, interactive tables, modal-based interactions, and a consistent design across all pages to improve user experience.

J. Backend Development

The backend was developed using Flask and Flask-SQLAlchemy. Major functionalities include REST API endpoints, authentication and role-based access control, data validation, error handling, and a modular route structure for better maintainability.

K. Database Design

An SQLite database was used to manage users, workers, jobs, applications, transactions, and reviews. Proper relational structure was maintained to ensure smooth data flow and reliable information retrieval across the system.

L. System Integration and Testing

All modules were integrated and tested to ensure a smooth workflow from job posting to review submission. Testing confirmed proper role-based access, accurate AI matching results, and reliable error handling and validation mechanisms.

IV. Tools and Technologies Used

The development of SafeHire involved several tools and technologies, including Python with Flask for backend development, Flask-SQLAlchemy for database operations, SQLite for data storage, HTML and Tailwind CSS for frontend design, JavaScript with Fetch API for interactive features, GitHub for version control, and Visual Studio Code as the primary development environment.

V. System Workflow

The complete workflow of the SafeHire platform begins when a worker creates a profile. The admin then verifies the worker before the employer posts a job. The worker receives AI-matched job recommendations and applies for suitable jobs. The employer reviews the applicants, hires a worker, and completes payment through the simulated transaction system. Finally, the employer submits a review, while the admin continues to monitor transactions and overall platform activity.

VI. Conclusion

The SafeHire platform successfully implements a complete service hiring system with AI-assisted matching, worker verification, and transparent transaction tracking. The system demonstrates strong backend architecture, clean and consistent user interface design, efficient workflow integration, and practical application of AI logic. It is suitable for academic evaluation and demonstrates real-world applicability.

VII. Future Improvements

Future improvements of the SafeHire platform may include real payment gateway integration, more advanced machine learning models for recommendation, a real-time chat system, notification services, mobile application development, and cloud deployment for wider accessibility and scalability.

VIII. Final Remark

SafeHire represents a complete solution for modern service hiring by combining AI assistance, verification mechanisms, and transparent workflows to build trust in digital marketplaces.